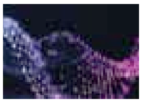
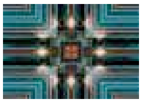


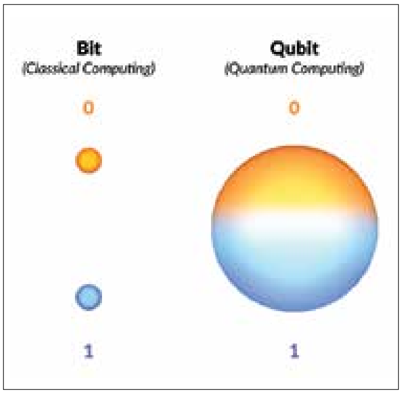
**Entangled photons**
Quantum encryption works using a pair of entangled photons that cannot be read without disrupton. **https://dgit.in/jul20-03**



**Quantum networking**
From diamonds to qubits, all the interesting experiments working towards building a quantum internet. **https://dgit.in/jul20-04**

state. Ideally, quantum networks will do more than just QKD and the next step is actually transferring quantum states or data.

However, there are some limitations since the data or qubits are still transferred over cables. Photons can be absorbed in the atmosphere or even by the materials in cables, which




Bit vs. Qubit

accounts for some data loss. This also means that the qubits can typically travel no more than a few kilometers. In 2017, China demonstrated some applications of QKD where quantum keys were transmitted from a satellite named Micius to two ground stations – one in Beijing and the other in Vienna. Any attempt to intercept this communication would effectively destroy them, making it seemingly impenetrable. However, they still ran into the issue of the travel limitations of qubits. This was apparently 'fixed' by employing 32 'trusted nodes' at various points along the network. These 'trusted nodes' are similar to traditional repeaters that are used to amplify the signal in data cables.

At these nodes, the encrypted keys have to be decrypted back to their traditional form and then re-crypted again to go back into quantum state until they reach the next node, where the they have to be decrypted and re-crypted again, and so on, and so forth. However, these nodes are actually entry-points for hackers, companies or even governments, where they could copy the keys without being detected, since the data here is in the traditional state and not quantum state. So, despite there being various combinations of classical and quantum internet that are already in use around the world, there's a need for a more robust method of transmission of qubits over long distances.

## A MORE REFINED APPROACH

A research centre called QuTech began working towards eliminating the aforementioned limitations to build a foolproof quantum internet. The process used by the researchers strongly relies on a phenomenon known as quantum entanglement. In this phenomenon, pairs of qubits (typically, photons of light) are merged together in a single quantum state. So, even if they stray away from each other, in totally opposite directions, they will maintain the quantum connection between them. Also, altering the state of one photon will instantaneously change the state of the other one too, no matter how far apart they are. So, if a pair of entangled photos is shared between two locations, the quantum information will get across due to the entangled nature of the particles. This process is known as quantum teleportation.

Entangled matter nodes can replace classical nodes in quantum networks, in that manner two people can share data without actually transmitting a qubit physically since the entangled state causes changes in photons. Nevertheless, to actually accomplish this across long distances, the entanglement needs to first be distributed throughout the quantum network. This is typically done via standard fibre-optic networks but now the nodes will be made up of entangled photons, that are extremely secure compared to traditional nodes. In January 2019, Ben Lanyon from the Institute for Quantum Optics and Quantum Information in Austria, along with his team set a record (at the time) for creating entanglement between matter and light spanning 50 kilometers of optical fibre.

Now in 2020, a team of researchers led by Stephanie Wehner from QuTech, hailing from the Delft University of Technology, began building a fully quantum network that aims to connect four cities in the Netherlands. In the Delft Network, the data will be transmitted using quantum techniques from end to end. Wehner's team previously showed off that their network can transmit data over about 1.5 kilometers. However, advancements in their research have caused them to be quite confident that their network will link the cities of Delft and The Hague by the end of 2020. The team will be using an array of quantum repeaters to ensure an unbroken connection between the two cities which will extend the reach

Stephanie Wehner, QuTech

of the network. Wehner believes that a global quantum network can be achieved by the end of this decade. Despite this positive estimate, suitable hardware needs to be built for quantum networks to work well on a global scale. Sometimes quantum systems only function in very low temperatures and most times they need to be kept from interacting with the environment. There are many hurdles to overcome before the quantum internet can become a stable add-on or upgrade to the traditional internet networks we have today. It's hard to give an accurate estimate of when exactly the quantum internet will be up and running the way it is intended to.